

Paramjeet Singh

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of production experience building LLM pipelines, agentic systems, and automated evaluation workflows. Strong intuition for model failure modes including hallucination patterns, prompt misinterpretation, and shortcut behaviors in frontier models. Experienced designing grading logic, directing AI agents through multi-step reasoning tasks, and iterating on task design based on where and why models fail. Independent by default; comfortable owning a problem end-to-end.

EDUCATION

Northeastern University

Master of Science, Information Systems

Grade : 3.72

Boston, MA

April 2026

CMR University

BTech, Electronics and Communication

Bangalore, India

Jun 2021

SKILLS

Languages: Python, SQL, JavaScript, Java

LLM & AI Engineering: Large Language Models (LLMs), GPT-4, Azure OpenAI, Prompt Engineering, Adversarial Prompt Design, Model Failure Analysis, Hallucination Detection, Retrieval-Augmented Generation (RAG), Agentic AI, Tool-Calling Pipelines, LangChain, Coding Agent Direction, Automated Grading & Validation, MindStudio

ML & Data: PyTorch, Pandas, NumPy, Scikit-learn, Vector Embeddings, Similarity Search, Dataset Curation

Backend & APIs: FastAPI, Flask, REST APIs, Microservices, Asynchronous Processing

Data Engineering: ETL Pipelines, Azure Data Factory, Databricks, SQL Server, MongoDB

Cloud & Infra: Azure OpenAI, Azure AI Search, Azure Blob Storage, Azure Functions, Docker, CI/CD

PROFESSIONAL EXPERIENCE

Acuvate Software

Bangalore, India

AI Engineer– Data & AI Solutions

Mar 2023 – Aug 2024

- Designed automated evaluation loops and grading logic for LLM classification pipelines, identifying systematic failure patterns, misclassification under edge-case inputs, retrieval hallucinations, prompt misinterpretation and iterating on prompt structure and grader design to reduce error rates by 60%
- Built production LLM pipelines using Azure OpenAI and GPT-4, developing adversarial test cases to expose model shortcut behaviors and validate robustness across varied input distributions
- Developed a Retrieval-Augmented Generation agent using GPT-4, Azure AI Search, and vector embeddings; analyzed model failure modes in multilingual retrieval scenarios to improve response accuracy and reduce hallucinated citations
- Implemented LLM orchestration pipelines with tool-calling and REST API integrations, validating agent decision logic and catching subtle reasoning errors before production deployment
- Built a complaint classification system integrating Azure OpenAI, OCR, and Azure Functions; automated grading of model outputs against ground-truth labels, reducing manual review workload by 60%

Acuvate Software

Hyderabad, India

Software Engineer – Data Engineering & Automation

Jun 2021 – Mar 2023

- Built scalable Python-based data ingestion pipelines across 100+ external sources, generating structured datasets used to train and evaluate NLP models
- Engineered a semantic search platform using embeddings and sentiment analysis, validating retrieval quality through automated scoring and reducing information retrieval time by 35%
- Designed and optimized SQL Server ETL pipelines with indexing strategies and stored procedures, improving data processing reliability by 50%

Northeastern University

Boston, MA

Instructional Assistant

Aug 2025 – Feb 2026

- Supported instruction for database design and SQL systems; reviewed and debugged student workflows, reinforcing best practices in schema design and query optimization

PROJECTS

CoachMe - AI Coaching System | Hackathon Winner | [GitHub](#)

- Won competitive AI hackathon by building an end-to-end video coaching system using 512-dimensional Marengo 3.0 embeddings for technique similarity scoring; designed 8 custom FiftyOne operators covering the full pipeline ingest, embed, compare, coach, validate, and curate
- Built a hallucination detection layer that cross-verifies every AI-generated coaching claim from Pegasus 1.2 against video evidence, directly tackling the problem of frontier models producing confident but unverifiable outputs

ConNET - AI Networking Agent | [GitHub](#)

- Built a multi-agent networking system using LangChain and GPT-4o orchestrating email, phone, and memory tools through a unified pipeline
- Implemented agent-to-agent communication protocol enabling autonomous exchange of professional context, mutual interest identification, and connection proposals without human intervention

QuickQuiz - Agentic AI Application | [GitHub](#)

- Built an agentic quiz generation system with automated grading and validation pipelines to catch subtle LLM failures, hallucinated answers, malformed structures, and inconsistent reasoning chains
- Directed LLM agents through multi-step reasoning tasks; developed failure recovery logic by analyzing how models misinterpret ambiguous prompts, improving answer accuracy by 35%